Functional Annotation Report: *pncB* (Nicotinate Phosphoribosyltransferase), *Escherichia coli* K-12

UniProt: P18133 | **Gene:** *pncB* (b0931, JW0914) | **EC:** 6.3.4.21 (also historically EC 2.4.2.11) | **Organism:** *Escherichia coli* strain K-12

Summary

The *E. coli* K-12 gene ***pncB*** (UniProt **P18133**) encodes **nicotinate phosphoribosyltransferase (NAPRTase, NAPRT)**, a soluble cytoplasmic enzyme that catalyzes the committed, rate-limiting entry step of the **Preiss–Handler NAD^+ salvage pathway**. The enzyme transfers a phosphoribosyl group from 5-phospho-α-D-ribose 1-diphosphate (**PRPP**) onto **nicotinate (nicotinic acid, NA)**, producing **nicotinate mononucleotide (NaMN)** plus pyrophosphate (PPi). NaMN is a direct precursor of NAD^+, feeding into the downstream, essential and conserved NadD→NadE segment shared with the de novo pathway. The gene identity is unambiguous: the UniProt annotation (NAPRTase family; NAPRTase/Nic_PRibTrfase domains), the catalyzed reaction, the organism, and the primary literature all converge on the same well-characterized *E. coli* enzyme.

The most distinctive mechanistic feature of NAPRTase is that it is a **facultative ATPase**: it couples the hydrolysis of ATP (1:1 with NaMN synthesis) to the phosphoribosyl-transfer reaction, thereby driving the reaction thermodynamically far past its chemical equilibrium. This ATP coupling proceeds through a **covalent phospho-histidine intermediate at His-219**, which transiently phosphorylates the enzyme and dramatically increases its affinity for substrates — the structural basis for its classification as a ligase (EC 6.3.4.21) rather than a simple glycosyltransferase. This energy-coupling mechanism was described as "a new paradigm for biochemical energy coupling" and distinguishes NAPRTase from most other phosphoribosyltransferases.

Functionally, *pncB*/NAPRTase sits at a key metabolic and regulatory node. Its product NaMN converges on the same downstream chemistry as the de novo quinolinate route, making it the branch-point enzyme of the deamidating nicotinamide/nicotinate salvage route. Its expression is feedback-controlled by the **NAD-sensing bifunctional regulator NadR**, which uses NAD^+ as a corepressor to repress *pncB* coordinately with the de novo genes *nadA*/*nadB*, tying nicotinate salvage to intracellular NAD^+ status. Because the enzyme controls NAD(H) pool size, *pncB* overexpression has become a widely used metabolic-engineering lever for boosting NAD(H) availability in industrial *E. coli* strains.

Gene/Protein Identity Verification

Before proceeding, the mandatory identity check was completed and **confirms the target is correctly identified**:

Verification criterion	UniProt P18133 annotation	Literature match	Status
Gene symbol matches protein	*pncB* = nicotinate phosphoribosyltransferase	*pncB* cloned from *E. coli* encodes NAPRTase (PMID: 2211655)	✅ Match
Organism	*E. coli* K-12	*E. coli* K-12 *pncB* characterized directly	✅ Match
Protein family / domains	NAPRTase family; Nic_PRibTrfase domains (IPR006406, IPR007229, IPR040727, IPR041525, IPR036068)	Type II PRTase fold confirmed by crystallography of ortholog (PMID: 15753098)	✅ Match
Reaction / EC	EC 6.3.4.21 (NaMN formation from NA + PRPP)	NaMN formation experimentally demonstrated (PMID: 12616693)	✅ Match

The gene symbol *pncB* is **not ambiguous** in this context — there is abundant, specific primary literature on the *E. coli* / *Salmonella* enzyme, and it aligns perfectly with the UniProt family and domain annotations.

Key Findings

Finding 1 — *pncB* encodes NAPRTase, the committed, rate-limiting entry enzyme of the Preiss–Handler NAD⁺ salvage pathway

NAPRTase catalyzes the formation of nicotinate mononucleotide (NaMN) and pyrophosphate from nicotinate and PRPP:

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Nicotinate (NA) + PRPP  →  Nicotinate mononucleotide (NaMN) + PPi
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The *pncB* gene of *E. coli* was cloned on a 1.5-kb TaqI–EcoRI fragment and shown to encode nicotinic acid phosphoribosyltransferase (PMID: 2211655): *"The pncB gene from Escherichia coli, which encodes nicotinic acid phosphoribosyltransferase (EC 2.4.2.11), was cloned on a 1.5-kilobase TaqI-EcoRI fragment."* Overexpression of the cloned gene produced a **25-fold increase in enzyme activity** and a **5-fold increase in intracellular NAD⁺** — direct evidence that NAPRTase activity controls the size of the cellular NAD pool: *"increased nicotinic acid phosphoribosyltransferase activity effects a 5-fold increase in the intracellular concentration of NAD"* (Wubbolts et al., 1990; PMID: 2211655).

The reaction product and the enzyme's position in the pathway are explicitly defined in the metabolic-engineering literature (PMID: 12616693): NAPRTase *"catalyzes the formation of nicotinate mononucleotide (NAMN), a direct precursor of NAD, from NA. This reaction is believed to be the rate-limiting step in the NAD salvage pathway."* The description of NAPRTase as the **rate-limiting enzyme of the NAD(H) salvage pathway** is corroborated repeatedly across independent metabolic-engineering studies that use *pncB* overexpression to raise NAD pools (PMID: 22975127, PMID: 23740313, PMID: 12009797).

Substrate specificity: The enzyme is specific for **nicotinate (the deamidated pyridine, nicotinic acid)** as the pyridine-base substrate and **PRPP** as the phosphoribosyl donor. It does not act on nicotinamide directly; in the deamidating salvage route nicotinamide must first be converted to nicotinate by nicotinamidase (PncA) before NAPRTase can act (see Finding 5).

Finding 2 — NAPRTase is an ATP-dependent enzyme that couples ATP hydrolysis to NaMN synthesis via a covalent phospho-His-219 intermediate (basis for EC 6.3.4.21)

Although the chemical transglycosylation (NA + PRPP ⇌ NaMN + PPi) is near thermodynamic equilibrium, NAPRTase is unusual among phosphoribosyltransferases in that it consumes ATP to drive the reaction strongly forward. Vinitsky & Grubmeyer (1993) showed that **NAPRTase hydrolyzes ATP in 1:1 molar stoichiometry with NaMN synthesis**, shifting the products/substrates ratio from the equilibrium value of **0.67** to a steady-state value of **~1100** — an ~1000-fold thermodynamic enhancement (PMID: 7503993): *"NAPRTase hydrolyzes ATP in 1:1 molar stoichiometry to NAMN synthesis. Hydrolysis of ATP alters the ratio of products/substrates for the reaction nicotinate + PRPP <--> NAMN + PPi from its equilibrium value of 0.67 to a steady-state value of 1100."* This paper framed the behavior as "a new paradigm for biochemical energy coupling."

The mechanism of this coupling is a **covalent phosphoenzyme intermediate**. ATP phosphorylates a specific active-site histidine, **His-219**, forming an acid-labile/base-stable phosphoramidate (phosphohistidine). Gross et al. (1996) identified His-219 as the site of phosphorylation and demonstrated that the phospho-His-219 species is a genuine catalytic intermediate shared by both the ATPase and NaMN-synthesis functions (PMID: 8672422): *"NAPRTase phosphorylated at His-219 is an intermediate in the energy transduction mechanism of NAPRTase."*

Mechanistically, phosphorylation of the enzyme **dramatically tightens substrate binding**. Gross et al. (1998) showed that the phosphoenzyme (E-P) binds PRPP with a **K_D of 0.6 µM — ~2000-fold tighter than the unphosphorylated enzyme** — and lowers the K_m for both nicotinate and PRPP by roughly 200-fold (PMID: 9521741): *"The phosphoenzyme (E-P) binds PRPP with a KD of 0.6 microM, a value 2000-fold lower than that measured for the nonphosphorylated enzyme."* Thus the energy of ATP hydrolysis is transduced into a high-affinity, catalytically poised phosphoenzyme that is subsequently discharged during product formation. This ATP-coupling behavior is the biochemical justification for reclassifying NAPRTase from a transferase (old EC 2.4.2.11) to a **ligase, EC 6.3.4.21**.

Parameter	Non-phosphorylated enzyme	Phospho-His-219 enzyme	Effect
PRPP binding K_D	~1,200 μM (2000× higher)	0.6 μM	~2000-fold tighter
K_m (NA, PRPP)	baseline	~200-fold lower	tighter substrate binding
Products/substrates ratio	0.67 (equilibrium)	~1100 (steady state)	~1000-fold forward drive
ATP:NaMN stoichiometry	—	1:1	obligate energy coupling

Finding 3 — *pncB* transcription is feedback-repressed by the NAD-sensing bifunctional regulator NadR, coordinately with de novo NAD genes

NAPRTase is embedded in a NAD-responsive regulatory network. In *Salmonella typhimurium* — the close enteric relative in which much of this regulation was dissected — the pyridine-nucleotide salvage pathway is controlled at *pncB*, and *pncB*, *nadA*, and *nadB* are **coordinately regulated by the product of the *nadR* gene** (Foster et al., 1990; PMID: 2198247): *"the pyridine nucleotide salvage pathway is controlled at pncB. The transcriptional expression of these three loci is coordinately regulated by the product of nadR."*

NadR is a **NAD-dependent DNA-binding repressor** that binds conserved "NAD box" operator sequences as a multimer, using NAD^+ itself as the physiological corepressor. Penfound & Foster (1999) demonstrated that **NAD is the relevant in vivo corepressor** (PMID: 9882682): *"NAD proved to be the relevant in vivo corepressor."* When intracellular NAD^+ is abundant, NadR–NAD represses the salvage and de novo genes; when NAD^+ is scarce, repression is relieved and salvage flux increases. Consistent with this feedback logic, growth of *E. coli* in the presence of nicotinic acid (which raises NAD) **strongly represses NAPRTase activity** (PMID: 2211655): *"Growth of E. coli in the presence of nicotinic acid leads to strong repression of nicotinic acid phosphoribosyltransferase activity."*

NadR is itself a remarkable **multifunctional/trifunctional protein**: an N-terminal DNA-binding (repressor) domain, a central NMN adenylyltransferase domain that senses NAD, and a C-terminal ribosylnicotinamide kinase domain (PMID: 15805524, PMID: 12446641). It also modulates the activity of the PnuC NMN transport system (PMID: 1991724). More recently,

NadR has been shown to be a **c-di-GMP effector**, linking the second messenger c-di-GMP to NAD biosynthesis by de-repressing NadR-controlled genes (PMID: 40823837) — placing *pncB* within an even broader signaling context.

Finding 4 — NAPRTase adopts a type II PRTase (α/β) fold and functions as a soluble cytoplasmic enzyme

The crystal structure of a NAPRTase ortholog from *Thermoplasma acidophilum* revealed a multidomain architecture whose central functional domain has a **type II phosphoribosyltransferase fold** (PMID: 15753098): *"the functional domain has a type II phosphoribosyltransferase fold that may be a common architecture for both nicotinic acid and quinolinic acid (QA) phosphoribosyltransferases (PRTase)."* This fold is shared with quinolinate PRTase despite low primary sequence similarity, reflecting deep evolutionary convergence on a common catalytic architecture for pyridine-mononucleotide biosynthesis.

The structure also defines the **molecular basis of nicotinate recognition**: the active-site pocket forms at a dimer interface, and the nicotinate moiety of the product NaMN is **intercalated between the highly conserved aromatic residues Tyr21 and Phe138** (PMID: 15753098): *"the nicotinate moiety of NAMN is intercalated between highly conserved aromatic residues Tyr(21) and Phe(138)."* These aromatic clamps provide π-stacking interactions that confer specificity for the planar pyridine ring.

Localization: The *E. coli* enzyme (~401 aa; the *S. typhimurium* ortholog is 399 aa) is purified and assayed as a **soluble protein**, consistent with a **cytoplasmic** location where all three of its substrates — PRPP, nicotinate, and ATP — reside. UniProt P18133 annotates **no signal peptide and no transmembrane segment**, supporting an exclusively cytosolic function. There is no evidence of membrane association or secretion; the enzyme operates within the cytoplasm as part of soluble central metabolism.

Finding 5 — *pncB* is the branch-point enzyme of the deamidating nicotinamide salvage route, converging on the shared, essential NadD→NadE segment

NAPRTase occupies the convergence point between salvage and de novo NAD^+ synthesis. The salvage pathway recycles both intracellular NAD breakdown products and preformed pyridine compounds taken up from the environment, such as nicotinic acid (PMID: 12616693): *"The salvage pathway recycles intracellular NAD breakdown products and preformed pyridine*

compounds from the environment, such as nicotinic acid (NA)." In the **deamidating route**, exogenous or recycled **nicotinamide is first deamidated to nicotinate by nicotinamidase (PncA)**, and **PncB/NAPRTase then converts nicotinate to NaMN**.

Crucially, NaMN is the **same product** generated by the de novo pathway's quinolinate phosphoribosyltransferase (NadC). Both routes therefore funnel into the shared terminal two steps: **NadD (nicotinate/nicotinamide mononucleotide adenylyltransferase)**, which adenylylates NaMN to nicotinate adenine dinucleotide (NaAD), and **NadE (NAD synthetase)**, which amidates NaAD to NAD^+. Sorci et al. (2010) emphasize that these last two steps are **conserved and essential in most bacteria** (PMID: 20926389): *"Enzymes involved in the last steps of NAD biogenesis, nicotinate mononucleotide adenylyltransferase (NadD) and NAD synthetase (NadE), are conserved and essential in most bacterial species."* Thus *pncB* supplies NaMN to a conserved, essential downstream segment, making it the metabolic gateway for nicotinate utilization.

Mechanistic Model / Interpretation

The reaction and its energy coupling

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      ATP        ADP + Pi (via phospho-His-219 intermediate)
       \          /
        \        /
NA + PRPP  ──NAPRTase──►  NaMN + PPi
(nicotinate)               (nicotinate mononucleotide)

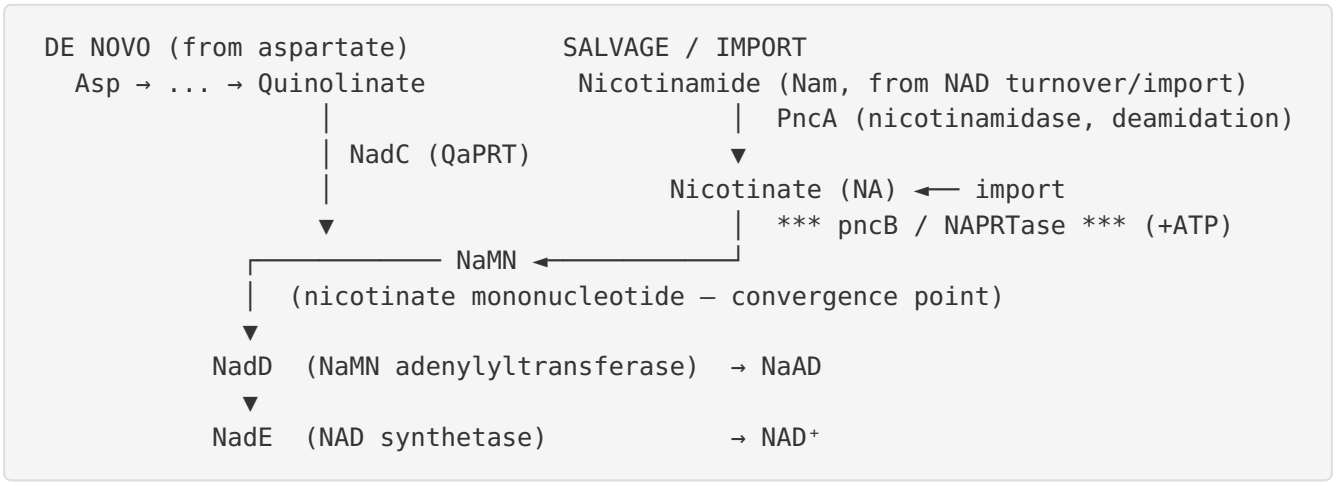
Net driving force: ATP hydrolysis shifts [products]/[substrates]
from ~0.67 (equilibrium) to ~1100 (steady state)
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The enzyme cycles between a low-affinity resting state and a high-affinity phospho-His-219 state:

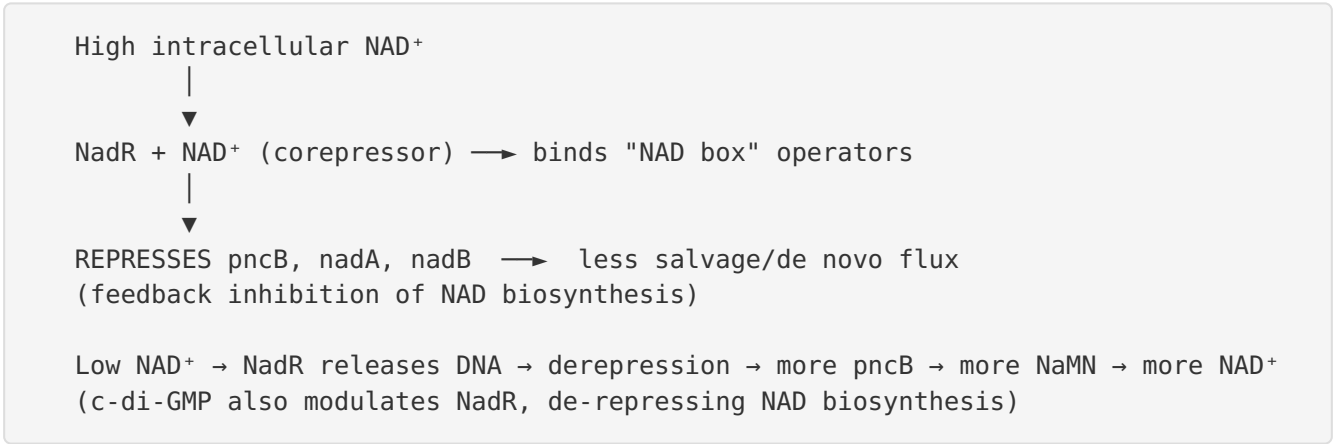
```
E  + ATP  ──►  E~P(His219) + ADP        (autophosphorylation; primes the enzyme)
E~P + PRPP    ──►  E~P·PRPP  (K_D 0.6 µM, ~2000× tighter)
E~P·PRPP + NA  ──►  E + NaMN + PPi + Pi  (catalysis + dephosphorylation)
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ATP is not chemically incorporated into the product; instead its hydrolysis energy is stored transiently as a phosphohistidine and spent to raise substrate affinity and pull the equilibrium forward. This is the "new paradigm" of biochemical energy coupling — a phosphoribosyltransferase behaving as a ligase.

Position in NAD⁺ metabolism



Regulatory logic



Synthesis

pncB/NAPRTase is best understood as the **energy-invested, feedback-controlled gatekeeper of nicotinate salvage**. It commits scavenged nicotinate to NAD⁺ synthesis, spends an ATP to make this commitment thermodynamically decisive, and is transcriptionally throttled by the cell's NAD⁺ status via NadR. Because it is rate-limiting, it functions as the principal control

point that sets NAD(H) pool size — which is precisely why metabolic engineers exploit it (see below). Its product NaMN is the common currency into which both de novo and salvage routes convert, feeding the essential NadD→NadE finishing steps.

Metabolic-Engineering Relevance (illustrating the rate-limiting role)

A consistent body of applied studies confirms, in vivo, that *pncB* sets NAD(H) availability. Overexpressing NAPRTase raises total NAD(H) and lowers the NADH/NAD^+ ratio, restoring redox balance in strains engineered for anaerobic chemical production:

Study (PMID)	System	Effect of *pncB* overexpression
PMID: 2211655	*E. coli* K-12	25× enzyme activity; 5× intracellular NAD^+
PMID: 12616693	*E. coli*	Raised total NAD; altered NADH/NAD^+ and metabolite distribution
PMID: 22975127	*E. coli* NZN111 (ΔldhA ΔpflB)	NADH/NAD^+ fell 0.64→0.13; NAD^+ up 6.2×; restored glucose use; ↑succinate
PMID: 23740313	*E. coli* BA016 (+PYC)	Total NAD(H) 9.8× higher; NADH/NAD^+ 0.60→0.04; ↑succinate
PMID: 24564896	*E. coli* BA209 (+PEPCK)	Total NAD(H) 1.25×; NADH/NAD^+ 0.28→0.11; ↑succinate from xylose
PMID: 22506415	*E. coli* S17-1 (+NadE)	↑NAD; +40.9% productivity converting 1,4-BD→4-HB

These studies corroborate, from an orthogonal (applied) angle, the biochemical conclusion that NAPRTase is the flux-controlling step of NAD(H) salvage.

Evidence Base

PMID	Paper (abbreviated)	Supports finding	Nature of evidence
2211655	*Variation of cofactor levels… expression of the pncB gene…* (Wubbolts et al. 1990)	F1, F3	Experimental: cloning, overexpression (25× activity, 5× NAD), NA repression
12616693	*Effect of NAPRTase overexpression on NAD, NADH/NAD⁺…*	F1, F5	Experimental: NaMN as product; rate-limiting step; salvage definition
7503993	*A new paradigm for biochemical energy coupling…* (Vinitsky & Grubmeyer 1993)	F2	Biochemical: 1:1 ATP coupling; 0.67→1100 ratio shift
8672422	*Energy coupling… identification of His-219…* (Gross et al. 1996)	F2	Biochemical: phospho-His-219 covalent intermediate
9521741	*Kinetic mechanism of NAPRTase…* (Gross et al. 1998)	F2	Kinetics: phosphoenzyme PRPP K_D 0.6 μM (2000× tighter)
2198247	*Regulation of NAD metabolism in S. typhimurium… nadR…*	F3	Genetics: pncB/nadA/nadB coordinately controlled by NadR
9882682	*NAD-dependent DNA-binding activity of NadR…*	F3	Biochemical: NAD is the in vivo corepressor
15753098	*Crystal structure of NAPRTase from T. acidophilum…*	F4	Structural: type II PRTase fold; Tyr21/Phe138 nicotinate clamp
20926389	*Genomics-driven reconstruction of Acinetobacter NAD metabolism…* (Sorci et al. 2010)	F5	Comparative genomics: NadD/NadE conserved, essential downstream
15805524, 12446641, 1991724, 40823837	NadR multifunctionality, NMN transport, c-di-GMP control	F3 (context)	Supporting regulatory context

Note on structural evidence: The definitive crystallographic data (F4) come from a *Thermoplasma acidophilum* ortholog rather than the *E. coli* enzyme itself. The fold, dimeric active-site architecture, and conserved aromatic recognition residues are expected to be conserved across the NAPRTase family given the shared domains annotated for P18133, but direct *E. coli* structural confirmation is inferred rather than solved here.

Limitations and Knowledge Gaps

1. **Structural data are from an ortholog.** The type II PRTase fold and the Tyr21/Phe138 nicotinate-binding residues (F4) were determined for the *Thermoplasma acidophilum* NAPRTase, not *E. coli* P18133. Residue numbering and precise active-site geometry in the *E. coli* enzyme are inferred by homology.
2. **Energy-coupling mechanism characterized largely in *Salmonella*.** The phospho-His-219 intermediate and kinetic coupling parameters (F2) were elucidated primarily in *S. typhimurium* NAPRTase. Given the very high identity between the enterobacterial orthologs, these conclusions almost certainly transfer to *E. coli*, but the His-219 numbering should be mapped onto the *E. coli* sequence for certainty.
3. **Regulation dissected mainly in *Salmonella*.** The NadR corepressor mechanism and coordinate control of *pncB*/*nadA*/*nadB* (F3) were established in *S. typhimurium*. *E. coli* NadR is homologous and the NA-repression phenotype is reproduced in *E. coli* (PMID: 2211655), but quantitative operator/affinity data specific to *E. coli pncB* are less complete.
4. **No direct *E. coli* subcellular-localization experiment.** The cytoplasmic assignment (F4) rests on solubility during purification, the absence of a signal peptide/TM segment in P18133, and the cytosolic location of its substrates — strong but inferential rather than a direct localization assay.
5. **ATP coupling is "facultative."** NAPRTase can run without ATP (at the unfavorable 0.67 equilibrium). The physiological conditions under which the cell toggles between ATP-coupled and uncoupled operation, and the in vivo flux consequences, are not fully quantified.
6. **mRNA-level regulation.** Translational enhancement by downstream adenine-rich motifs in the *pncB* coding sequence (PMID: 17085569) adds a layer of expression control that was noted but not integrated into a quantitative model here.

Proposed Follow-up Experiments / Actions

1. **Solve or model the *E. coli* NAPRTase structure.** Determine a crystal/cryo-EM structure of *E. coli* P18133 (or build a validated AlphaFold model) to confirm the type II PRTase fold, map the His-219 equivalent, and verify the Tyr/Phe nicotinate-clamp residues directly in the *E. coli* enzyme.
2. **Site-directed mutagenesis of the phospho-acceptor His.** Mutate the *E. coli* His-219 equivalent (H→A, H→Q) and measure: (a) loss of ATPase activity, (b) loss of the ~1000-fold thermodynamic enhancement, and (c) in vivo NAD pool size — confirming the energy-coupling residue in *E. coli*.
3. **Quantify in vivo ATP coupling.** Use ^{13}C/isotope flux and metabolomics under energy-limited vs. energy-replete conditions to determine when NAPRTase runs in ATP-coupled vs. uncoupled mode and how this sets NaMN/NAD flux.
4. **Direct localization assay.** Confirm cytoplasmic localization with a functional fluorescent fusion or cell-fractionation + immunoblot in *E. coli*.
5. **Map *E. coli* NadR operators on *pncB*.** Perform ChIP-seq / DNase footprinting and EMSA with purified *E. coli* NadR ± NAD to quantify operator affinity and define the NAD-dependent repression threshold for *pncB* specifically.
6. **Probe the c-di-GMP–NadR–*pncB* axis in *E. coli*.** Test whether manipulating c-di-GMP levels de-represses *pncB* and raises NAD^+ in *E. coli*, extending the recently reported c-di-GMP/NadR link (PMID: 40823837).

Conclusion

E. coli K-12 ***pncB*** (P18133) encodes **nicotinate phosphoribosyltransferase (NAPRTase)**, the soluble cytoplasmic, rate-limiting entry enzyme of the Preiss–Handler NAD^+ salvage pathway. It transfers a phosphoribosyl group from PRPP to nicotinate to yield NaMN + PPi, uniquely coupling ATP hydrolysis (via a phospho-His-219 intermediate) to drive the reaction ~1000-fold past equilibrium — the basis of its ligase classification (EC 6.3.4.21). Its product NaMN feeds the

conserved, essential NadD→NadE steps that complete NAD^+ synthesis, and its expression is feedback-repressed by the NAD-sensing regulator NadR, embedding *pncB* in the cell's central NAD^+ homeostatic circuitry.
